WINGMAN

A volatility-smile trading agent for SPY options

Alpaca AI Trading Agents Hackathon · Hubert Kasinski · 04/09/26

BACKGROUND

A short history of pricing options

1. Black-Scholes-Merton
2. Know limitations
3. What came next

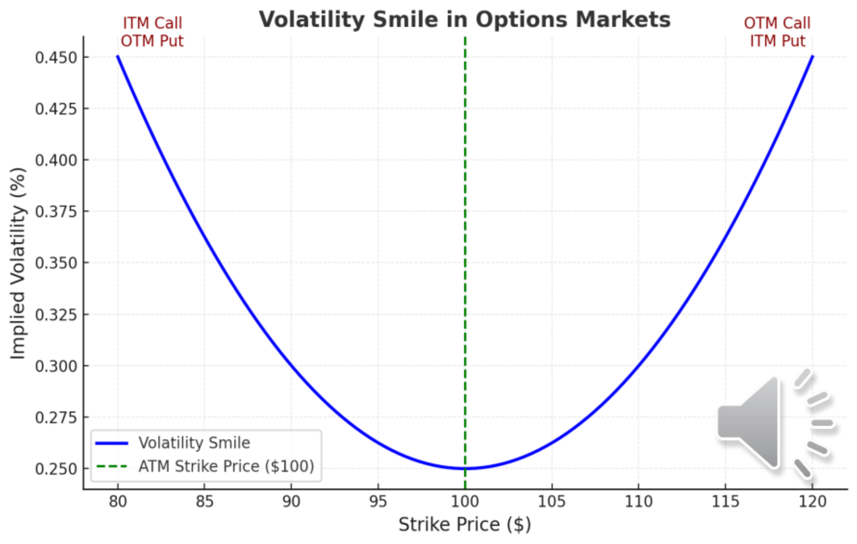
$$C = SN(d_1) - Ke^{-rt}N(d_2)$$

where:

$$d_1 = \frac{ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$$

and

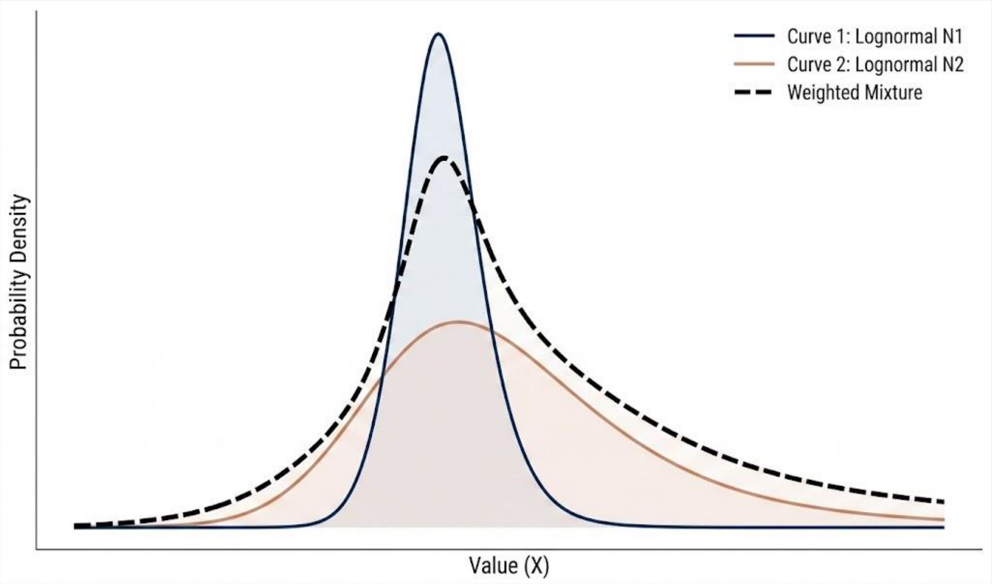
$$d_2 = d_1 - \sigma\sqrt{t}$$



Mixture Dynamics — two Black-Scholes worlds, blended

$$V_{mix}(K,T) = \lambda \cdot V_{BS}(S_0,K,T,\sigma_1,r,q) + (1-\lambda) \cdot V_{BS}(S_0,K,T,\sigma_2,r,q)$$

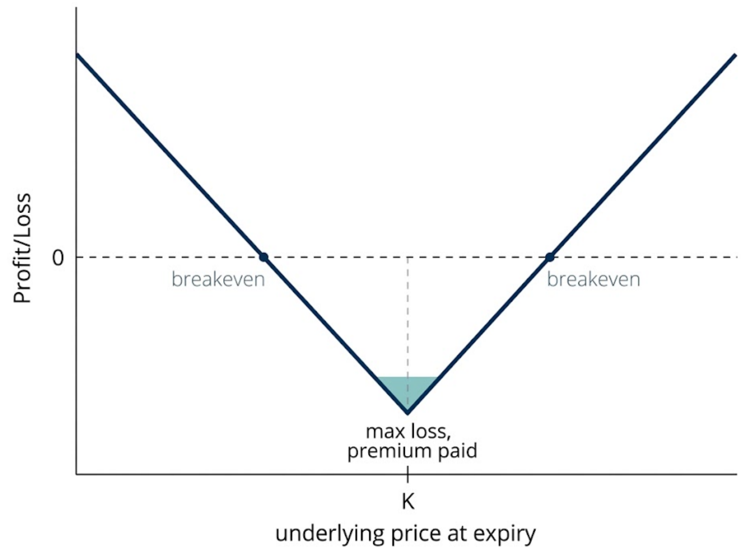
- **Two components**
- **Arbitrage-free by construction**
- **Refit every cycle**
- **Two components are enough**



Trading the gap, not the direction

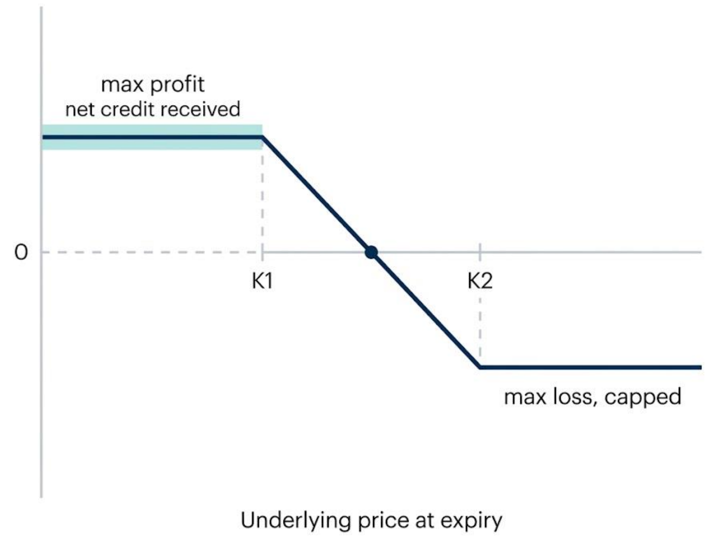
Market cheap vs model

→ LONG STRADDLE



Market rich vs model

→ SHORT CALL VERTICAL



Five independent checks before any order reaches the market

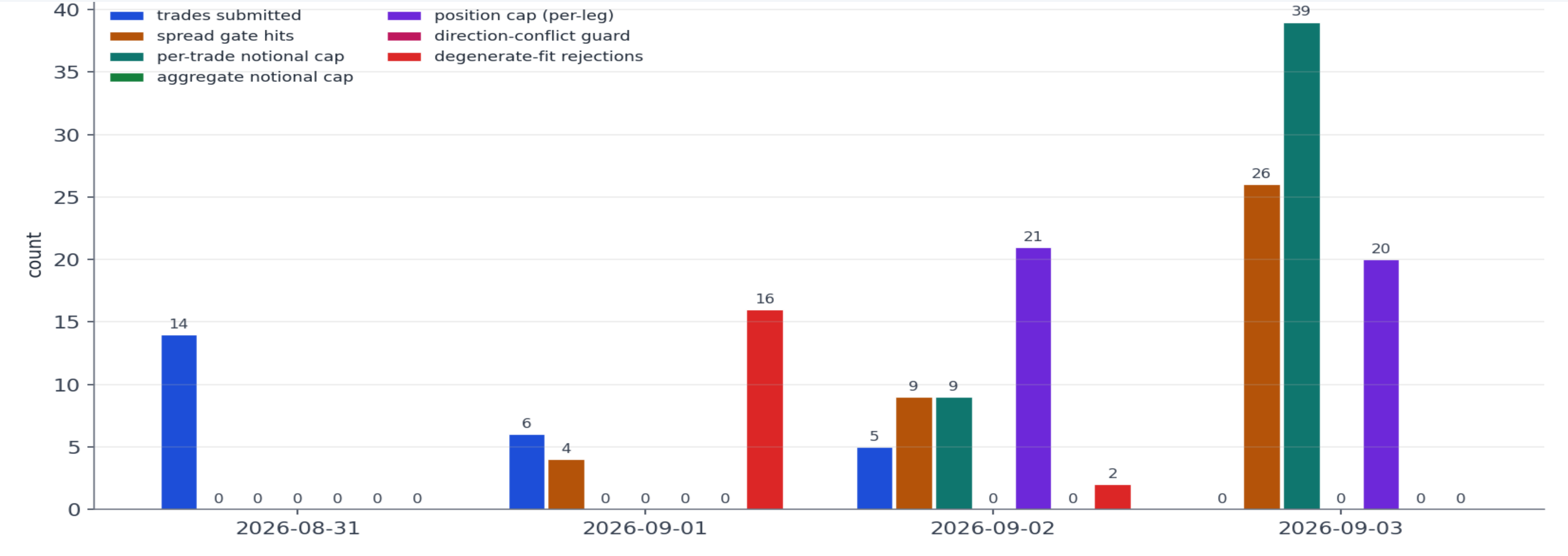
1. Half-spread gate
2. Notional cap
3. Position cap
4. Aggregate exposure cap
5. Direction-conflict guard

+ AI regime layer (Gemini)

FOUND LIVE, FIXED LIVE

What actually happened this week

The bug	The fix	The new find	The proof
MON	TUE	WED	THU



RESULTS

Four days, in numbers

110

cycles run

25

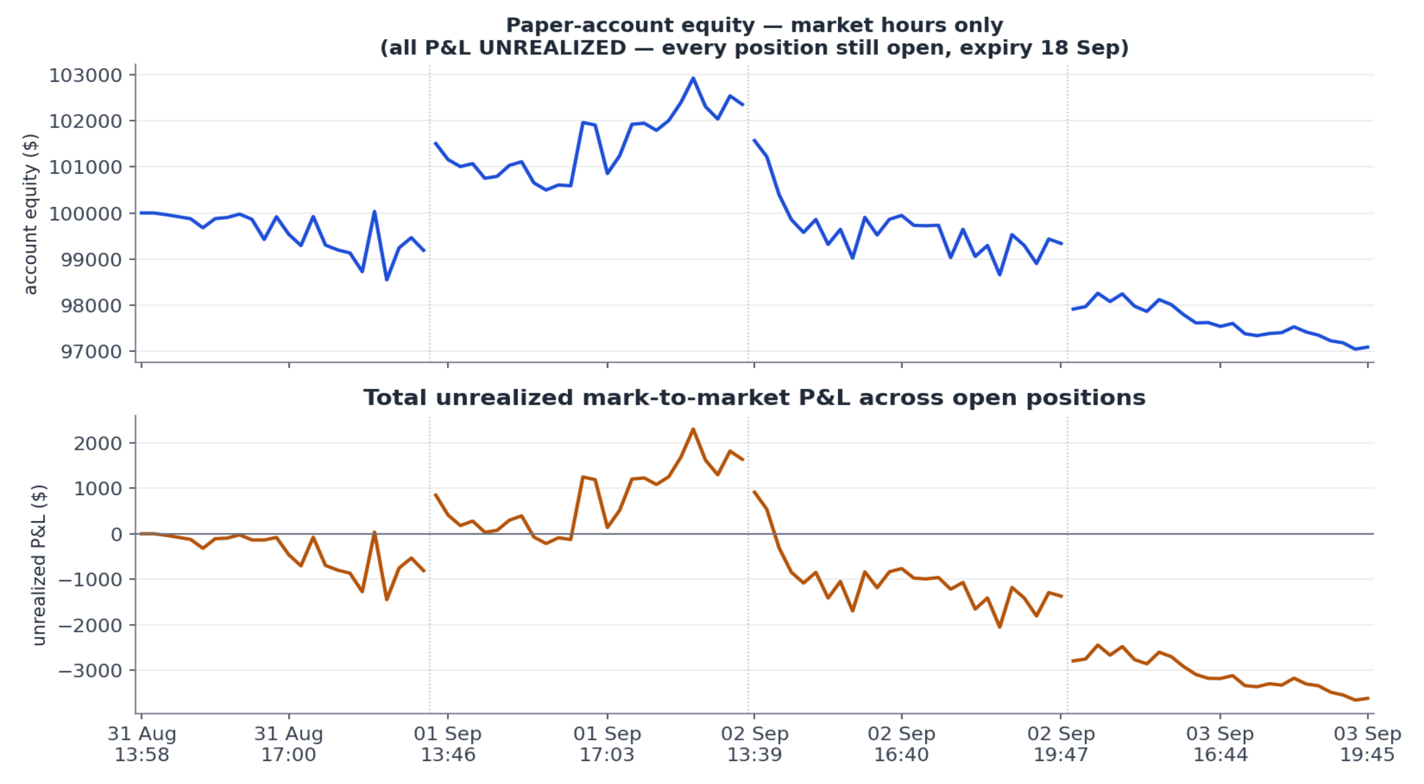
orders submitted, 20 filled

5

deterministic risk gates

–2.91%

week equity, mostly unrealized



What's still open

KNOWN LIMITATIONS

- Position cap is per-symbol, not per-cluster
- No portfolio-level long/short balancing
- Symmetric mixture vs. SPY’s real skew

FURTHER WORK

- Shift-extended mixture dynamics
- Cluster-aware exposure cap
- Multi-expiry calibration
- Faster cadence

WHERE ELSE THIS FITS

Beyond SPY

- Other liquid chains
- FX
- Institutional risk workflow

Thank you

Paper account: PA3ICB1OURR1

github.com/UbbaDubb/Wingman

Nearly all P&L is unrealized mark-to-market. One manual risk-reduction trade (1 Sep) realized +$714, closing 8 duplicated Monday positions before automated position caps existed.